LAB 8:

BANKERS ALGORITHM

```
#include <stdio.h>

void inputMatrices(int n, int m, int alloc[n][m], int max[n][m], int avail[m]) {
    printf("Enter Allocation Matrix (%d x %d):\n", n, m);
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++)
            scanf("%d", &alloc[i][j]);

    printf("Enter Max Matrix (%d x %d):\n", n, m);
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++)
            scanf("%d", &max[i][j]);

    printf("Enter Available Resources (%d values):\n", m);
    for (int i = 0; i < m; i++)
        scanf("%d", &avail[i]);
}

void calculateNeed(int n, int m, int alloc[n][m], int max[n][m], int need[n][m]) {
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++)
            need[i][j] = max[i][j] - alloc[i][j];
}

void bankersAlgorithm(int n, int m, int alloc[n][m], int need[n][m], int avail[m]) {
    int f[n], ans[n], ind = 0;
    for (int i = 0; i < n; i++) f[i] = 0;

    for (int k = 0; k < n; k++) {
        for (int i = 0; i < n; i++) {
            if (f[i] == 0) {
                int flag = 0;
                for (int j = 0; j < m; j++) {
                    if (need[i][j] > avail[j]) {
                        flag = 1;
                        break;
                    }
                }
                if (flag == 0) {
                    ans[ind++] = i;
                    for (int y = 0; y < m; y++) avail[y] += alloc[i][y];
                    f[i] = 1;
                }
            }
        }
    }
```

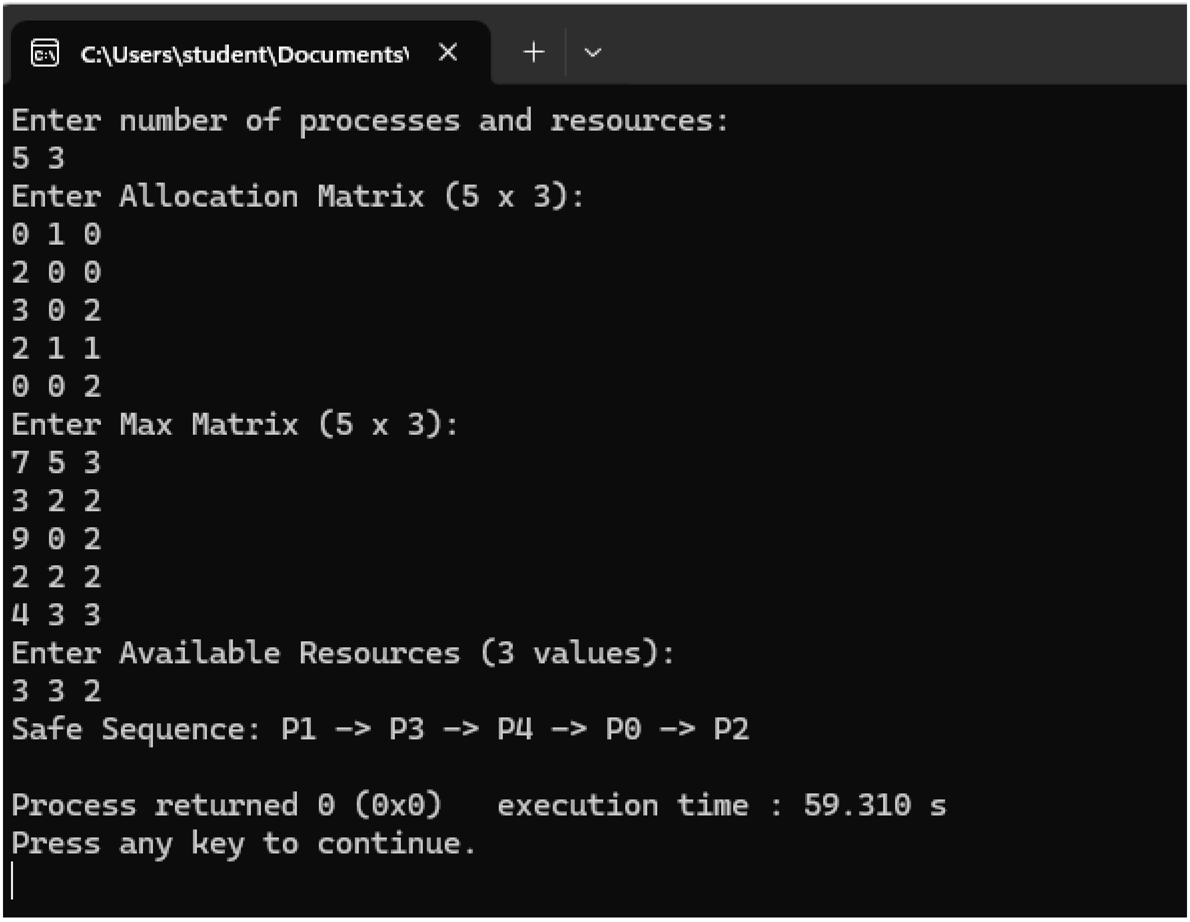
```
    printf("Safe Sequence: ");
    for (int i = 0; i < n - 1; i++) printf("P%d -> ", ans[i]);
    printf("P%d\n", ans[n - 1]);
}

int main() {
    int n, m;
    printf("Enter number of processes and resources:\n");
    scanf("%d %d", &n, &m);

    int alloc[n][m], max[n][m], avail[m], need[n][m];
    inputMatrices(n, m, alloc, max, avail);
    calculateNeed(n, m, alloc, max, need);
    bankersAlgorithm(n, m, alloc, need, avail);

    return 0;
}
```

OUTPUT:



C:\Users\student\Documents\

```
Enter number of processes and resources:
5 3
Enter Allocation Matrix (5 x 3):
0 1 0
2 0 0
3 0 2
2 1 1
0 0 2
Enter Max Matrix (5 x 3):
7 5 3
3 2 2
9 0 2
2 2 2
4 3 3
Enter Available Resources (3 values):
3 3 2
Safe Sequence: P1 -> P3 -> P4 -> P0 -> P2

Process returned 0 (0x0)   execution time : 59.310 s
Press any key to continue.
```